

UBT846: COMPUTATIONAL BIOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: To understand the basics of computational science and apply it to solve biological problems. To familiarize students with basic concepts of algorithms used in sequences, structural alignment, database searching, protein structure prediction, computer-based drug designing and machine learning classifiers for biological systems.

Syllabus

Introduction: Definition, Goals, applications of Computational Biology, Biological molecules such as DNA and Protein sequence, their structure and function. DNA and protein sequence databases. Structural databases

Basics of Computer Programming Problem solving Technique: Pseudocode, Algorithm, Flowchart, Data structures – Array, Stack, Queue, Linked, List concepts. Shell commands and scripting

Algorithms: Classification of algorithms, HMM profile searches, Algorithms to analyze system biology problems such as high throughput genomics and proteomics data, Machine-learning approaches, binary classification model development with examples in biological systems

Molecular computational biology: Gene prediction, sequencing genomes Protein structure determination by computational methods such as molecular modeling and molecular threading. Impact of the change of single amino acid (SNP) on the structure and function of the proteins, Protein-protein interactions, Functional insight by molecular dynamic simulations

Tutorials: Shell programming, Multiple sequence alignment, DNA binding motif finding by sequence alignment

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Perform programming Linux shell.
2. Explain various types of algorithms with their possible application in solving biological problems.
3. Explain the computational aspects of complex data analyses.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text Books

1. Pevzner, P. A., Computational Molecular Biology, PHI Learning Pvt. Ltd, ISBN-978-81-203-2550-0.
2. Ghosh, Z. and Mallick, B., Bioinformatics Principles and Applications (2008) Oxford University Press ISBN 9780195692303.

Reference Books

1. Mount D W, Bioinformatics - Sequence and Genome Analysis, Cold Spring Harbour Laboratory Press (2001), 2nd ed.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/ quizzes)	30

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